

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
df=pd.read_csv("/content/StudentPerformance.csv")
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 29 entries, 0 to 28
Data columns (total 6 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Maths_Score            29 non-null    int64
1   Reading_Score          29 non-null    int64
2   Writing_Score          29 non-null    int64
3   Placement_Score        29 non-null    int64
4   Club_Join_Date         29 non-null    int64
5   Placement offer count  29 non-null    int64
dtypes: int64(6)
memory usage: 1.5 KB
```

```
df.isnull().sum()
```

	0
Maths_Score	0
Reading_Score	0
Writing_Score	0
Placement_Score	0
Club_Join_Date	0
Placement offer count	0

dtype: int64

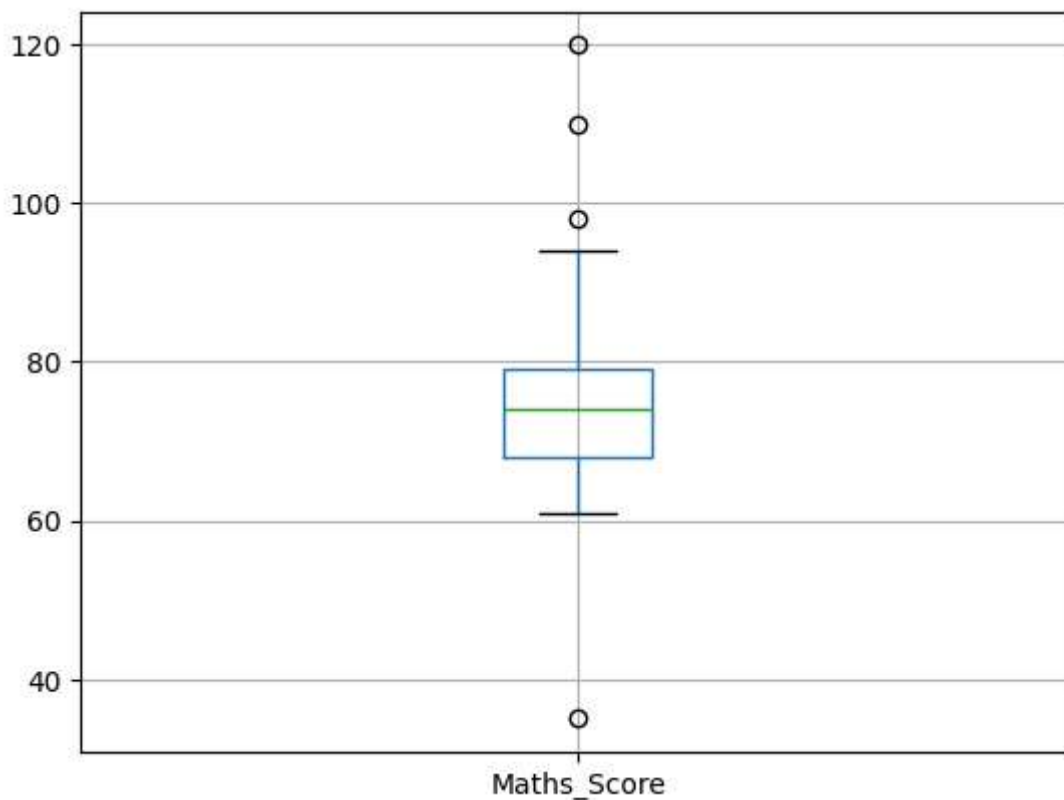
```
mean_val=df['Maths_Score'].mean()
df['Maths_Score']=df['Maths_Score'].fillna(mean_val)
mean_val=df['Reading_Score'].mean()
df['Reading_Score']=df['Reading_Score'].fillna(mean_val)
mean_val=df['Writing_Score'].mean()
df['Writing_Score']=df['Writing_Score'].fillna(mean_val)
mean_val=df['Placement_Score'].mean()
df['Placement_Score']=df['Placement_Score'].fillna(mean_val)
df.isnull().sum()
```

	0
Maths_Score	0
Reading_Score	0
Writing_Score	0
Placement_Score	0
Club_Join_Date	0
Placement offer count	0

dtype: int64

```
col=['Maths_Score']  
df.boxplot(col)
```

<Axes: >



```
sorted_df=sorted(df['Maths_Score'])  
sorted_df  
q1=np.percentile(sorted_df,25)  
q3=np.percentile(sorted_df,75)  
print(q1,q3)
```

68.0 79.0

```
IQR=q3-q1  
lwr_bound=q1-(1.5*IQR)  
upr_bound=q3+(1.5*IQR)  
print(lwr_bound,upr_bound)
```

```
51.5 95.5
```

```
sample_outliers=df[col][(df[col]<lwr_bound)|(df[col]>upr_bound)]  
sample_outliers
```

	Maths_Score	
0	NaN	
1	NaN	
2	NaN	
3	35.0	
4	NaN	
5	NaN	
6	NaN	
7	NaN	
8	NaN	
9	NaN	
10	NaN	
11	NaN	
12	NaN	
13	NaN	
14	NaN	
15	110.0	
16	NaN	
17	NaN	
18	NaN	
19	120.0	
20	NaN	
21	NaN	
22	NaN	
23	NaN	
24	NaN	
25	NaN	
26	NaN	
27	98.0	
28	NaN	

```

from sklearn.preprocessing import LabelEncoder
le = LabelEncoder ()
df['gender'] = le.fit_transform(df['gender'])
newdf=df
df

```

```

-----
KeyError                                Traceback (most recent call last)
/usr/local/lib/python3.12/dist-packages/pandas/core/indexes/base.py in
get_loc(self, key)
    3804         try:
-> 3805             return self._engine.get_loc(casted_key)
    3806         except KeyError as err:

index.pyx in pandas._libs.index.IndexEngine.get_loc()

index.pyx in pandas._libs.index.IndexEngine.get_loc()

pandas/_libs/hashtable_class_helper.pxi in
pandas._libs.hashtable.PyObjectHashTable.get_item()

pandas/_libs/hashtable_class_helper.pxi in
pandas._libs.hashtable.PyObjectHashTable.get_item()

KeyError: 'gender'

```

The above exception was the direct cause of the following exception:

```

KeyError                                Traceback (most recent call last)
----- 2 frames -----
/usr/local/lib/python3.12/dist-packages/pandas/core/indexes/base.py in
get_loc(self, key)
    3810         ):
    3811             raise InvalidIndexError(key)
-> 3812             raise KeyError(key) from err
    3813         except TypeError:
    3814             # If we have a listlike key, check indexing error will

```